# ECAS Summer Course

## Quantile Regression for Longitudinal Data

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Part I: Penalty Methods for "Random" Effects
Part II: Estimation of Reference Growth Charts

Part II is based on joint work with:
Ying Wei, Columbia, Biostatistics
Anneli Pere, Children's Hospital, Helsinki
and Xuming He, UIUC.

# Penalty Methods and "Random" Effects

- Classical Models with Independent Random Effects
  - ⋆ Additive random effects have nice conditional mean functions and nasty conditional quantile functions.
  - ⋆ Transformations to annihilate fixed effects fail for quantile regression models

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  - ⋆ Additive random effects have nice conditional mean functions and nasty conditional quantile functions.
  - ⋆ Transformations to annihilate fixed effects fail for quantile regression models

- Classical Models with Fixed Effects
  - ⋆ Fixed effect models are computationally tractable
  - ⋆ Penalty methods for fixed effect models offer an alternative to classical random effects estimation

# A Classical Result

**Model:**

$$y_{ij} = x_{ij}^\top \beta + \alpha_i + u_{ij} \quad j = 1, ...m_i, \quad i = 1, ..., n,$$

or,

$$y = X\beta + Z\alpha + u.$$

If $u$ and $\alpha$ are independent Gaussian vectors with $u \sim \mathcal{N}(0, R)$ and $\alpha \sim \mathcal{N}(0, Q)$, the optimal weighted least squares estimator is,

$$\hat{\beta} = (X^\top (R + ZQZ^\top)^{-1} X)^{-1} X^\top (R + ZQZ^\top)^{-1} y.$$

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**Theorem** $\hat{\beta} = \text{argmin}_{(\alpha,\beta)} \|y - X\beta - Z\alpha\|_{R^{-1}}^2 + \|\alpha\|_{Q^{-1}}^2$, where $\|x\|_A^2 = x^\top A x$.

# Commentary

**History** Henderson (1950) on dairy cows, Goldberger (1962) on BLUP, Lindley and Smith (1972) on Bayesian hierarchical models, and Robinson (1991) for a survey.

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**Interpretation** Maximum likelihood estimations for the Gaussian random effects models is identical to penalized least squares estimation of the fixed effects model. Random effects is just shrinkage estimation.

**Ideology** Our focus is on design of effective algorithms for data analysis. Models are useful as an interpretative device, but they shouldn't be allowed to become sacred objects.

# Quantile Regression with Fixed Effects

**Model:**

$$Q_{y_{ij}}(\tau|x_{ij}) = \alpha_i + x_{ij}^\top \beta(\tau) \quad j = 1, ...m_i, \quad i = 1, ..., n.$$

# Quantile Regression with Fixed Effects

**Model:**

$$Q_{y_{ij}}(\tau|x_{ij}) = \alpha_i + x_{ij}^\top \beta(\tau) \quad j = 1, ...m_i, \quad i = 1, ..., n.$$

**Estimator:** With (the inevitable) $\rho_\tau(u) = u(\tau - I(u < 0))$,

$$\min_{(\alpha,\beta)} \sum_{k=1}^{q} \sum_{j=1}^{n} \sum_{i=1}^{m_i} w_k \rho_{\tau_k}(y_{ij} - \alpha_i - x_{ij}^\top \beta(\tau_k))$$

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**Implementation:**

- Linear programming formulation
- Interior point algorithms exploiting sparse linear algebra

# Quantile Regression with Penalized Fixed Effects

**Model: (as before)**

$$Q_{y_{ij}}(\tau|x_{ij}) = \alpha_i + x_{ij}^\top \beta(\tau) \quad j = 1, ...m_i, \quad i = 1, ..., n.$$

**Estimator:**

$$\min_{(\alpha,\beta)} \sum_{k=1}^{q} \sum_{j=1}^{n} \sum_{i=1}^{m_i} w_k \rho_{\tau_k}(y_{ij} - \alpha_i - x_{ij}^\top \beta(\tau_k)) + \lambda \sum_{i=1}^{q} |\alpha_i|$$

**Limiting Forms:**

- as $\lambda \to 0$ we obtain fixed effects,
- as $\lambda \to \infty$ we force all $\alpha_i$ to 0,

# $L_1$ versus $L_2$ Shrinkage

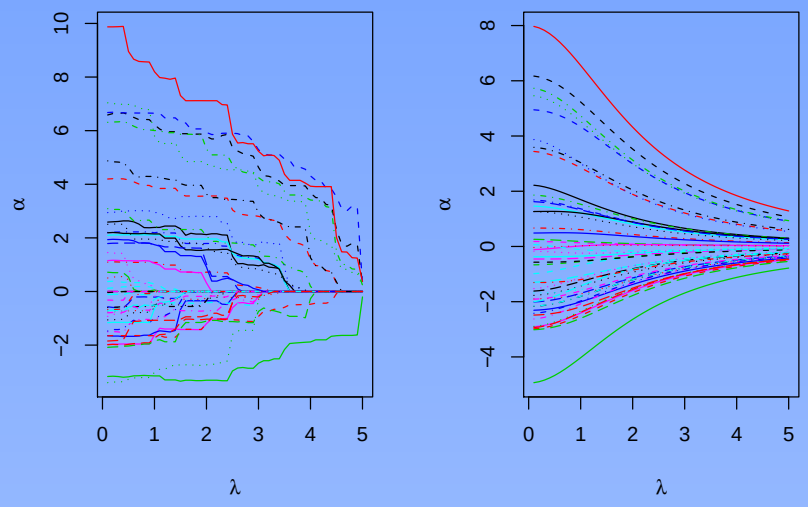


Figure 1: Shrinkage of the fixed effect parameter estimates, $\hat{\alpha}_i$. The left panel illustrates an example of the $\ell_1$ shrinkage effect. The right panel illustrates an example of the $\ell_2$ shrinkage effect. There are $m = 5$ observations on each of $n = 50$ cross-sectional units.

# Sparsity

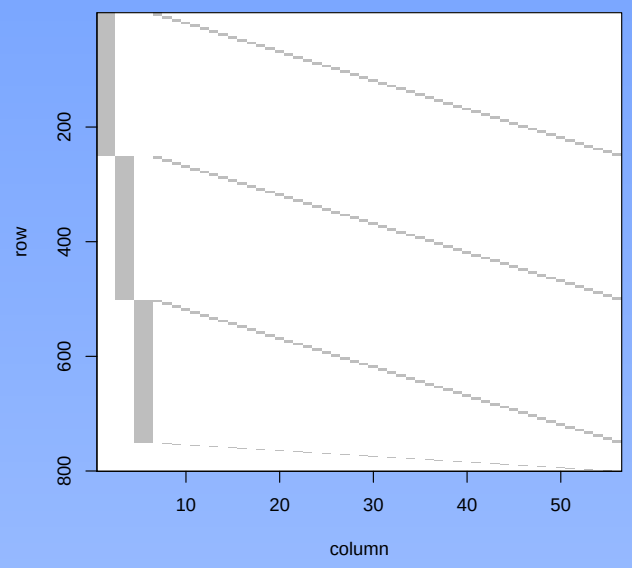


Figure 2: Sparsity of the "design" matrix for the penalized fixed effects quantile regression estimator. Grey areas indicate portions of the matrix that are nonzero, white space indicates zeros. Here, $n = 50$ and $m_i = 5$ for all $i$. Three quantiles are being simultaneously estimated with the fixed effects restricted to be identical across quantiles.

# Regularity Conditions

**A1.** The $y_{ij}$ are independent with conditional distribution functions, $F_{ij}$, given $x_{ij}$, and differentiable conditional densities, $0 < f_{ij} < \infty$, with bounded derivatives $f'_{ij}$, at $\xi_{ij}(\tau)$, for $j = 1, \ldots, m,\ i = 1, \ldots, n$,

**A2.** Let $\omega = \tau(1-\tau)$ and denote $\Phi = \text{diag}(f_{ij}(\xi_{ij}(\tau)))$. The limiting forms of the following matrices are positive definite:

$$D_0 = \lim_{\substack{m\to\infty \\ n\to\infty}} \frac{\omega}{m} \begin{pmatrix} Z^\top Z & Z^\top X/\sqrt{n} \\ X^\top Z/\sqrt{n} & X^\top X/n \end{pmatrix},$$

$$D_1 = \lim_{\substack{m\to\infty \\ n\to\infty}} m^{-1} \begin{pmatrix} Z^\top \Phi Z & Z^\top \Phi X/\sqrt{n} \\ X^\top \Phi Z/\sqrt{n} & X^\top \Phi X/n \end{pmatrix}.$$

**A3.** $\max_{\substack{1\le i\le n \\ 1\le j\le m}} ||x_{ij}|| < M$.

# A Theorem

**Theorem 1.** *Under conditions A1-3, with* $n^a/m \to 0$ *for some* $a > 0$*, the* $\hat{\delta}_1$ *component of the minimizer,* $\hat{\delta}$*, converges in distribution to a Gaussian random vector with mean zero and covariance matrix given by the lower* $p$ *by* $p$ *block of the matrix* $D_1^{-1} D_0 D_1^{-1}$*.*

# More Regularity Conditions

**B1.** The $y_{ij}$ are independent with conditional distribution functions, $F_{ij}$, given $x_{ij}$, and differentiable conditional densities, $0 < f_{ij} < \infty$, with bounded derivatives $f'_{ij}$, at $\xi_{ij}(\tau)$, for $j = 1, \ldots, m,\ i = 1, \ldots, n$,

**B2.** Let $\Omega$ denote the $q$ by $q$ matrix with typical element $\tau_k \wedge \tau_l - \tau_k \tau_l$ and $\Phi_k = \text{diag}(f_{ij}(\xi_{ij}(\tau_k)))$. The limiting forms of the following matrices are positive definite:

$$D_0 = \lim_{\substack{m\to\infty \\ n\to\infty}} m^{-1} \begin{pmatrix} w^\top \Omega w Z^\top Z & w^\top \Omega W \otimes Z^\top X/\sqrt{n} \\ W\Omega w \otimes X^\top Z/\sqrt{n} & W\Omega W \otimes X^\top X/n \end{pmatrix},$$

$$D_1 = \lim_{\substack{m\to\infty \\ n\to\infty}} m^{-1} \begin{pmatrix} \sum w_k Z^\top \Phi_k Z & w_1 Z^\top \Phi_1 X/\sqrt{n} & \cdots & w_q Z^\top \Phi_q X/\sqrt{n} \\ w_1 X^\top \Phi_1 Z/\sqrt{n} & w_1 X^\top \Phi_1 X/n & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ w_q X^\top \Phi_q Z/\sqrt{n} & 0 & \cdots & w_q X^\top \Phi_q X/n \end{pmatrix}.$$

**B3.** $\max_{\substack{1\leq i\leq n\\ 1\leq j\leq m}} ||x_{ij}|| < M.$

# Another Theorem

**Theorem 2.** *Under conditions B1-3, provided that* $\lambda_m/\sqrt{m} \to \lambda_0$, *and* $n^a/m \to 0$ *for some* $a > 0$, *the first component* $\hat{\delta}_1$ *minimizing* $V_{mn}$ *has the same limiting distribution as the as the first component of the minimizer of,*

$$V_0(\delta) = -\delta^\top B + \frac{1}{2}\delta^\top D_1 \delta + \lambda_0 \delta^\top s$$

*where* $B$ *is a zero mean Gaussian vector with covariance matrix* $D_0$, *and* $s = (s_0^\top 0_{pq}^\top)^\top$ *and* $s_0 = (sgn(\alpha_i))$.

# Monte-Carlo Models

We consider two very simple models:

$$y_{ij} = \alpha_i + x_{ij}\beta + u_{ij} \tag{1}$$

and

$$y_{ij} = \alpha_i + x_{ij}\beta + (1 + x_{ij}\gamma)u_{ij}. \tag{2}$$

where

$$x_{ij} = \delta_i + v_{ij} \tag{3}$$

with $\gamma_i$ and $v_{ij}$ independent and identically distributed over $i$ and $i, j$ respectively, then the interclass correlation coefficient,

$$\rho_x = \sigma_\delta^2/(\sigma_\delta^2 + \sigma_u^2),$$

# Monte Carlo: Location Shift Model

| | LS | PLS | LSFE | QR | PQR | QRFE |
|---|---|---|---|---|---|---|
| $\mathcal{N}$ | | | | | | |
| Bias | 0.0031 | 0.0048 | 0.0056 | 0.0048 | 0.0067 | 0.0047 |
| RMSE | 0.0847 | 0.0604 | 0.0668 | 0.0977 | 0.0781 | 0.0815 |
| $t_3$ | | | | | | |
| Bias | -0.0062 | -0.0054 | -0.0051 | -0.0063 | -0.0101 | -0.0082 |
| RMSE | 0.1377 | 0.1031 | 0.1143 | 0.1274 | 0.0881 | 0.0921 |
| $\chi^2_3$ | | | | | | |
| Bias | -0.0068 | 0.0002 | 0.0032 | -0.0052 | 0.0063 | 0.0072 |
| RMSE | 0.2155 | 0.1503 | 0.1650 | 0.2362 | 0.1506 | 0.1513 |

# Monte Carlo: Location-Scale Shift Model

| | LS | PLS | LSFE | QR | PQR | QRFE |
|---|---|---|---|---|---|---|
| $\mathcal{N}$ | | | | | | |
| Bias | 0.0000 | 0.0010 | 0.0012 | -0.0020 | -0.0021 | -0.0022 |
| RMSE | 0.0559 | 0.0501 | 0.0542 | 0.0638 | 0.0526 | 0.0556 |
| $t_3$ | | | | | | |
| Bias | -0.0045 | 0.0000 | 0.0008 | -0.0044 | -0.0015 | 0.0021 |
| RMSE | 0.0948 | 0.0806 | 0.0870 | 0.0758 | 0.0620 | 0.0693 |
| $\chi^2_3$ | | | | | | |
| Bias | 0.0617 | 0.0609 | 0.0608 | 0.0317 | -0.0055 | -0.0128 |
| RMSE | 0.1608 | 0.1292 | 0.1368 | 0.1627 | 0.1042 | 0.1092 |

# Conclusions

- Classical random effects in Gaussian models is just penalized least squares estimation in which fixed effects are shrunken toward their prior mean of zero. Other penalties may be useful in other settings.

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- $L_1$ penalty methods have attractive properties in a variety of model selection, image analysis and function estimation problems. Interior point algorithms exploiting inherently sparse linear algebra facilate efficient computations.

# Conclusions

- Classical random effects in Gaussian models is just penalized least squares estimation in which fixed effects are shrunken toward their prior mean of zero. Other penalties may be useful in other settings.

- $L_1$ penalty methods have attractive properties in a variety of model selection, image analysis and function estimation problems. Interior point algorithms exploiting inherently sparse linear algebra facilate efficient computations.

- There is more to longitudinal data analysis than the estimation of location shift effects. Many interesting questions can only be addressed with broader models that incorporate distributional effects for treatments and other covariates.

# Quantile Regression Methods for Reference Growth Charts

Roger Koenker
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Based on joint work with:
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Anneli Pere, Children's Hospital, Helsinki
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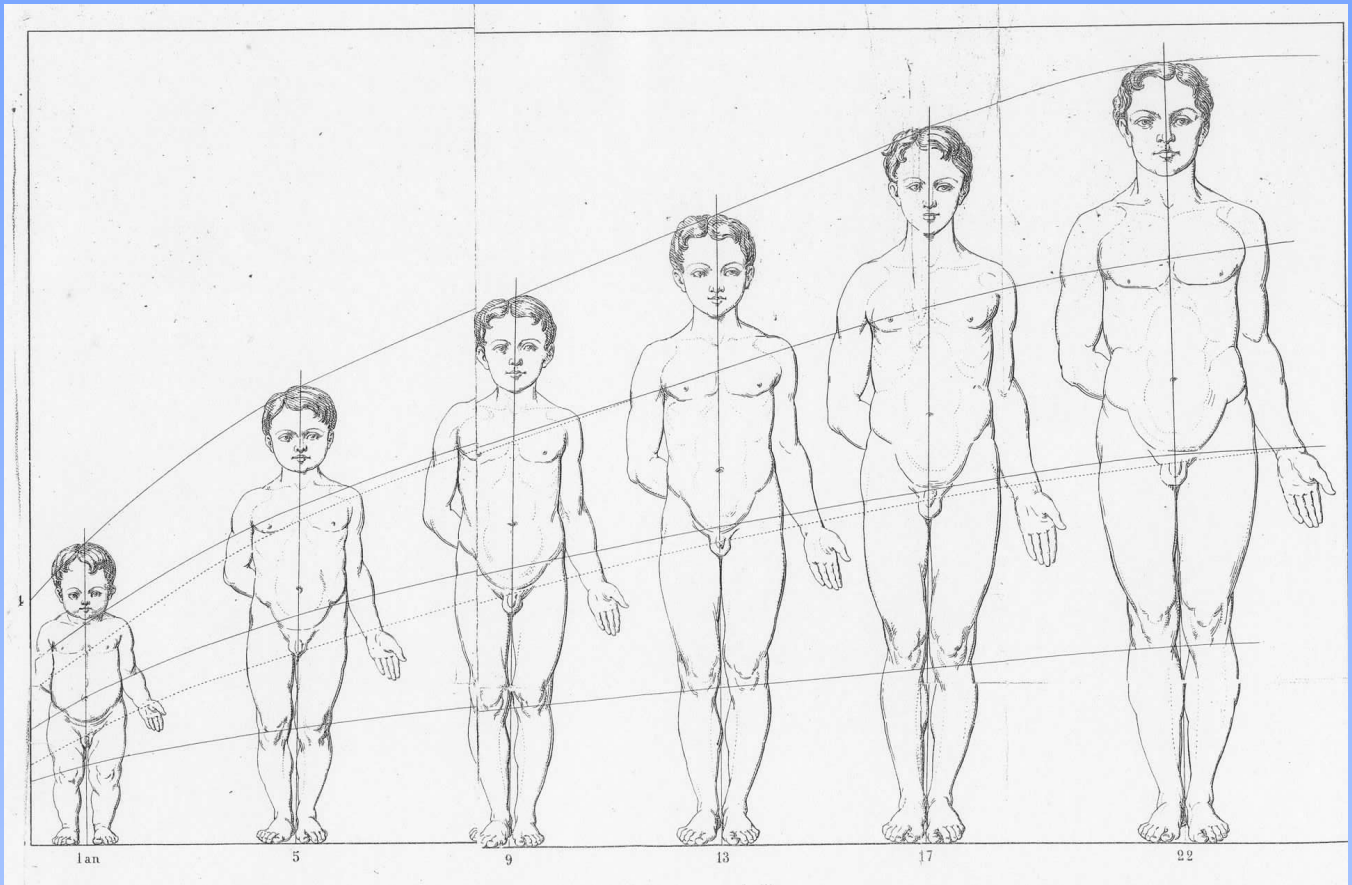
# Quetelet's (1871) Growth Chart



# Penalized Maximum Likelihood Estimation

- References: Cole (1988), Cole and Green (1992), and Carey(2002)
- Data: $\{Y_i(t_{i,j}) : \ j = 1, \ldots, J_i, \ i = 1, \ldots, n.\}$
- Model: $Z(t) = \frac{(Y(t)/\mu(t))^{\lambda(t)} - 1}{\lambda(t)\sigma(t)} \sim \mathcal{N}(0, 1)$
- Estimation:

$$\max \ell(\lambda, \mu, \sigma) - \nu_\lambda \int (\lambda''(t))^2 dt - \nu_\mu \int (\mu''(t))^2 dt - \nu_\sigma \int (\sigma''(t))^2 dt,$$

$$\ell(\lambda, \mu, \sigma) = \sum_{i=1}^{n} [\lambda(t_i) \log(Y(t_i)/\mu(t_i)) - \log \sigma(t_i) - \tfrac{1}{2} Z^2(t_i)],$$

# Quantiles as Argmins

The $\tau$th quantile of a random variable $Y$ having distribution function $F$ is:

$$\alpha(\tau) = \text{argmin} \int \rho_\tau(y - \alpha) dF(y)$$

where

$$\rho_\tau(u) = u \cdot (\tau - I(u < 0)).$$

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where

$$\rho_\tau(u) = u \cdot (\tau - I(u < 0)).$$

The $\tau$th *sample* quantile is thus:

$$\begin{aligned} \hat{\alpha}(\tau) &= \text{argmin} \int \rho_\tau(y - \alpha) dF_n(y) \\ &= \text{argmin}\ n^{-1} \sum_{i=1}^{n} \rho_\tau(y_i - \alpha) \end{aligned}$$

# Quantile Regression

The $\tau$th conditional quantile function of $Y|X = x$ is

$$g(\tau|x) = \text{argmin}_{g \in \mathcal{G}} \int \rho_\tau(y - g(x))dF$$

A natural estimator of $g(\tau|x)$ is

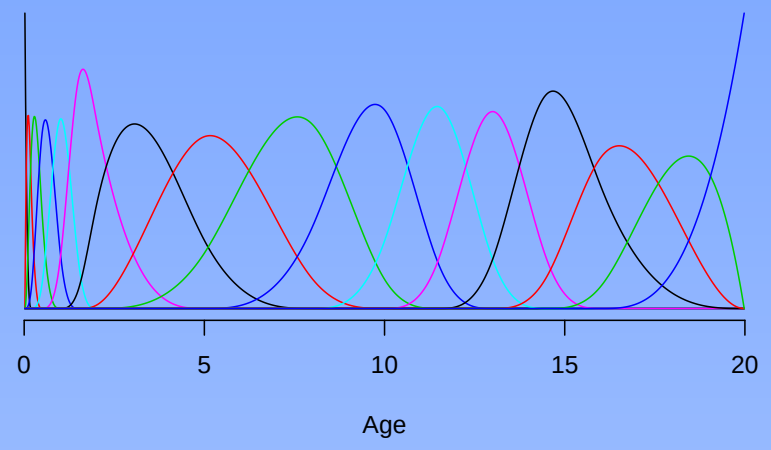
$$\hat{g}(\tau|x) = \text{argmin}_{g \in \mathcal{G}} \sum_{i=1}^{n} \rho_\tau(y_i - g(x_i))$$

with $\mathcal{G}$ chosen as a finite dimensional linear space,

$$g(x) = \sum_{j=1}^{p} \varphi_j(x)\beta_j.$$

# Choice of Basis

There are many possible choices for the basis expansion $\{\varphi_j\}$. We opt for the (very conventional) cubic B-spline functions:




In R these quantile regression models can be estimated with the command.

```
fit <- rq(y ~ bs(x,knots=knots),tau = 1:9/10)
```

Similar functionality in SAS is coming "real soon now."

# Data

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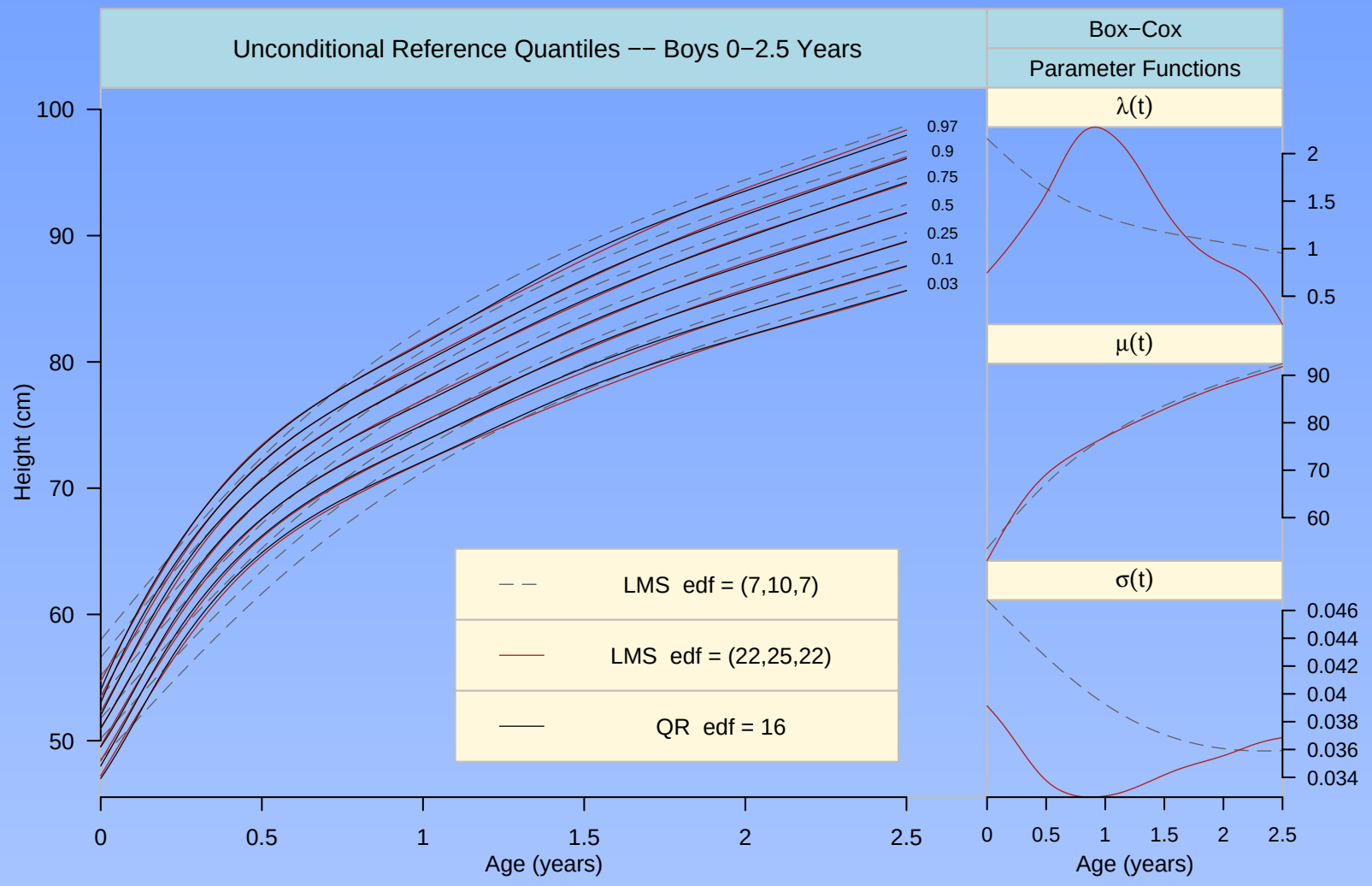
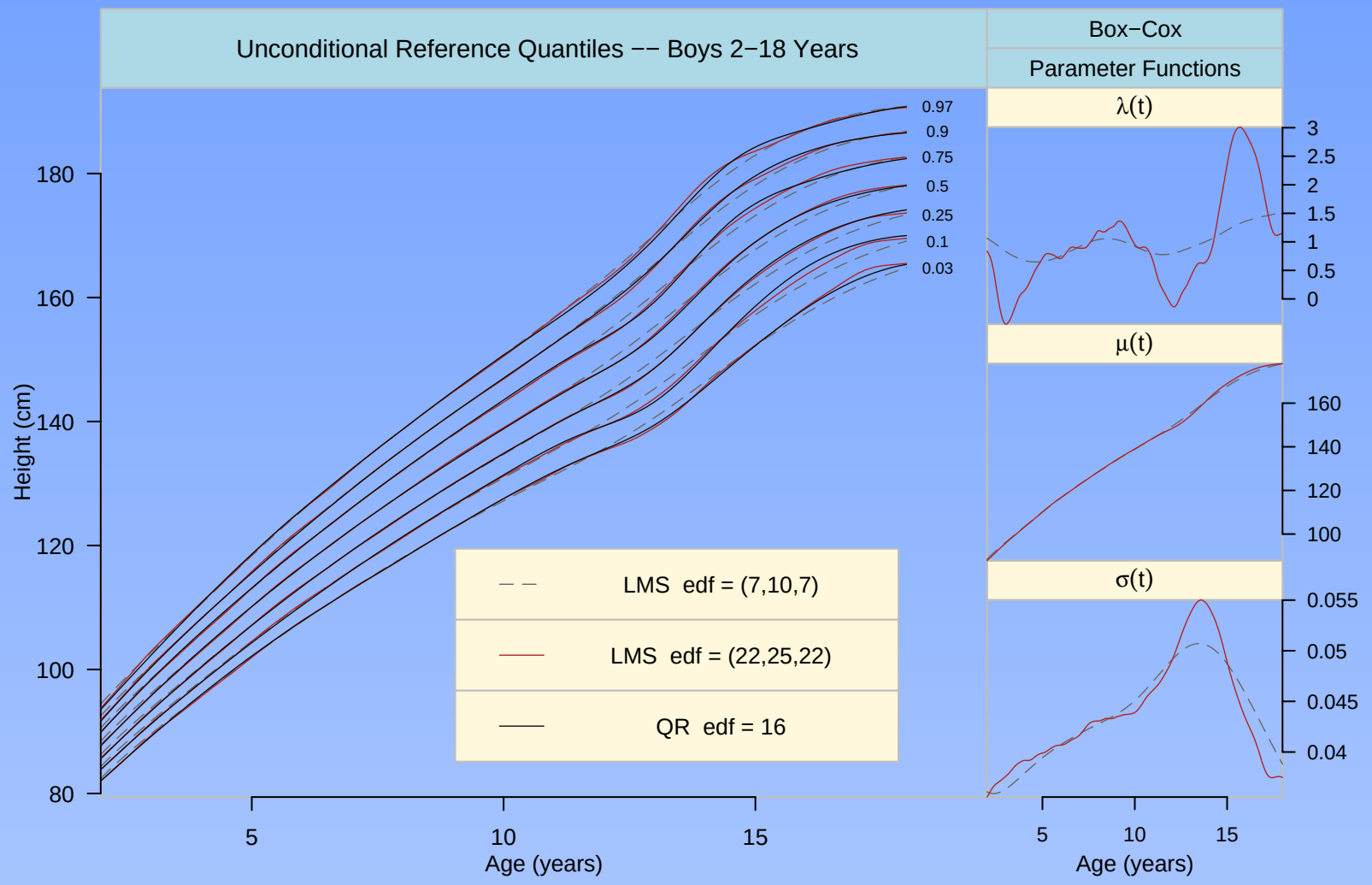
- 1143 boys, 1162 girls – all healthy, full-term, singleton births,
- About 20 measurements per child,
- Two cohorts: 1096 born between 1959-61, 1209 born between 1968-72
- Sample constitutes 0.5 percent of Finns born in these periods.


Unconditional Reference Quantiles −− Boys 0−2.5 Years
Box−Cox
Parameter Functions
λ(t)
μ(t)
σ(t)
Height (cm)
Age (years)
0.97
0.9
0.75
0.5
0.25
0.1
0.03
LMS edf = (7,10,7)
LMS edf = (22,25,22)
QR edf = 16


Unconditional Reference Quantiles −− Boys 2−18 Years
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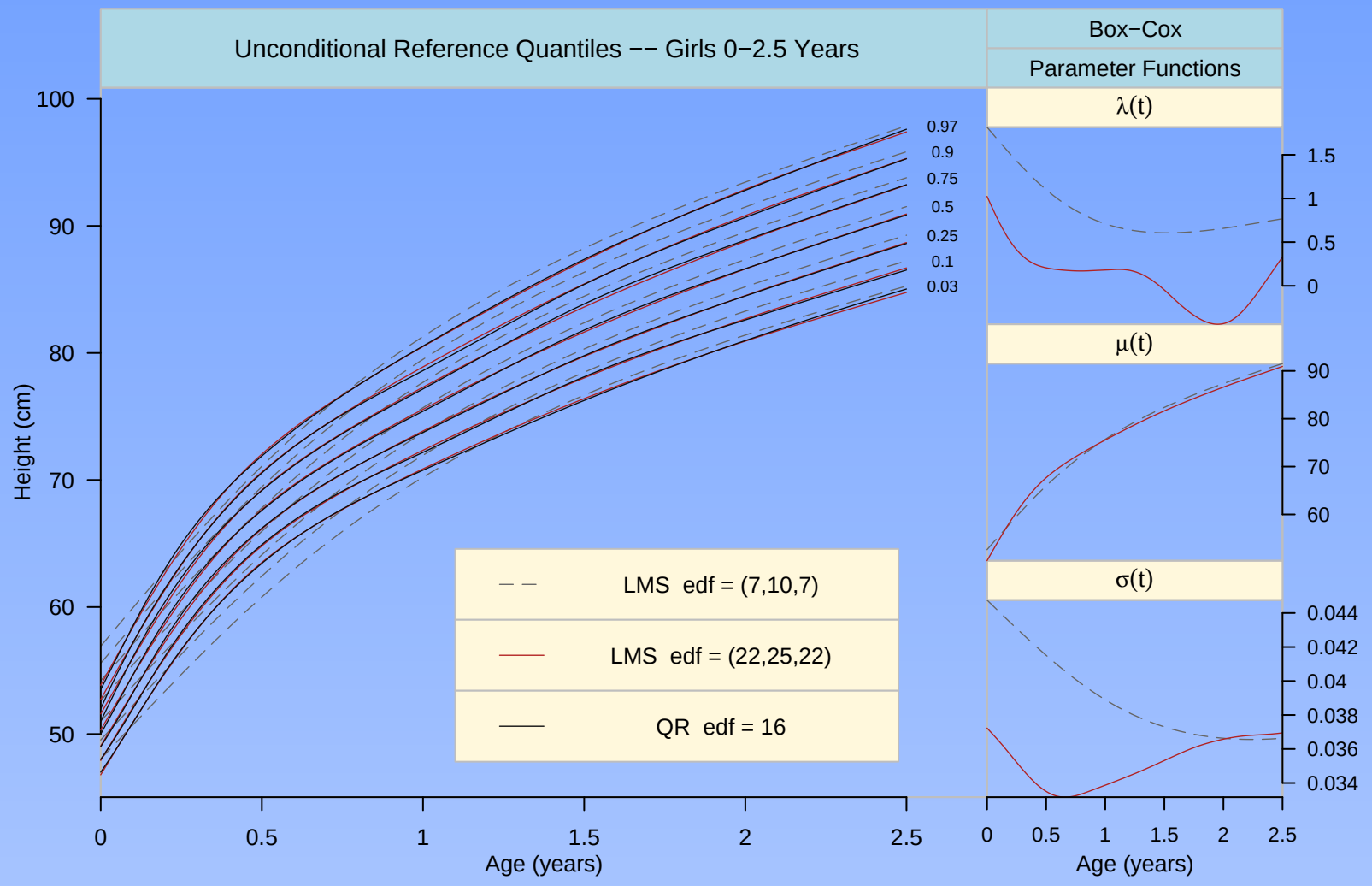
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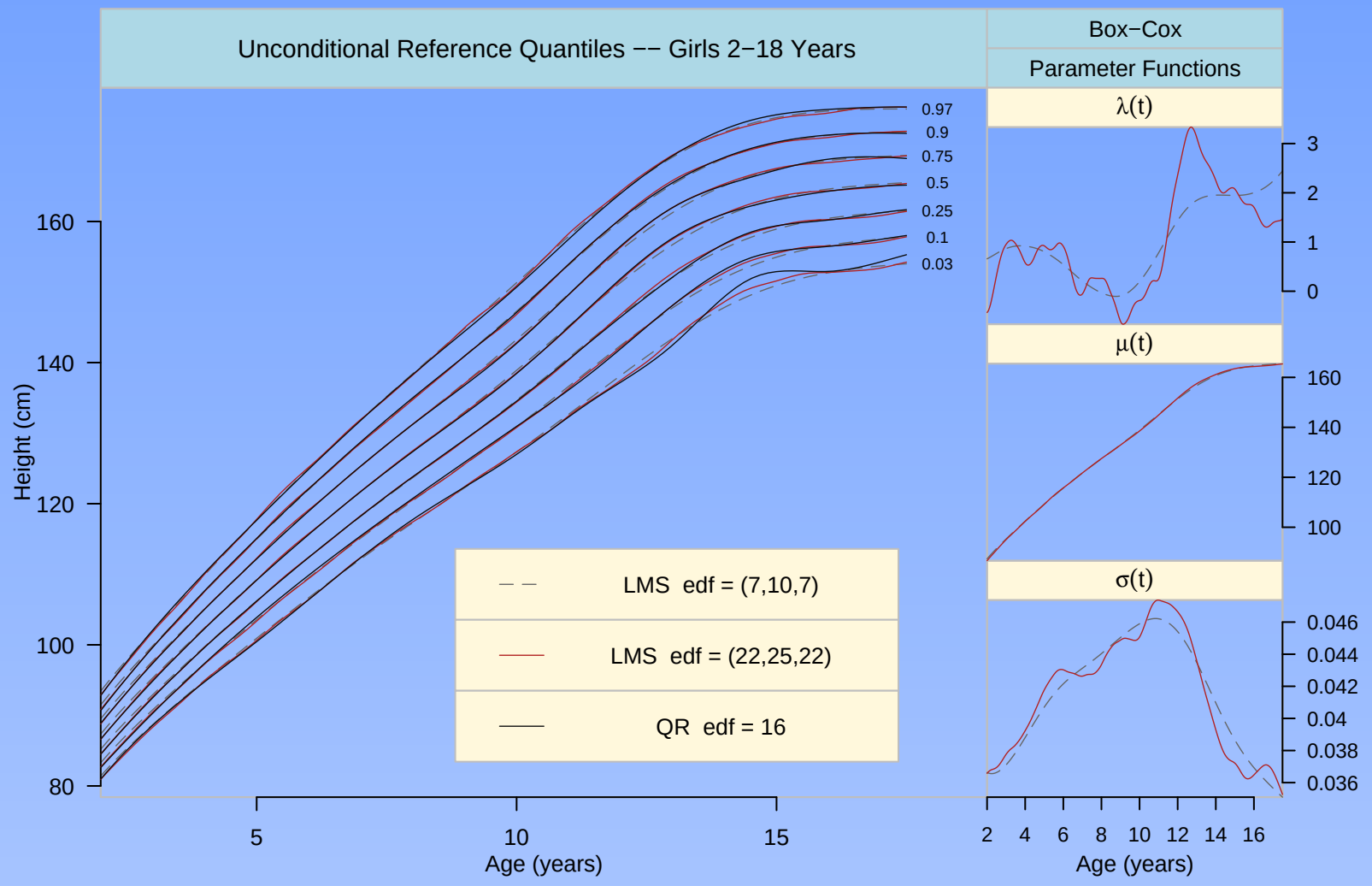
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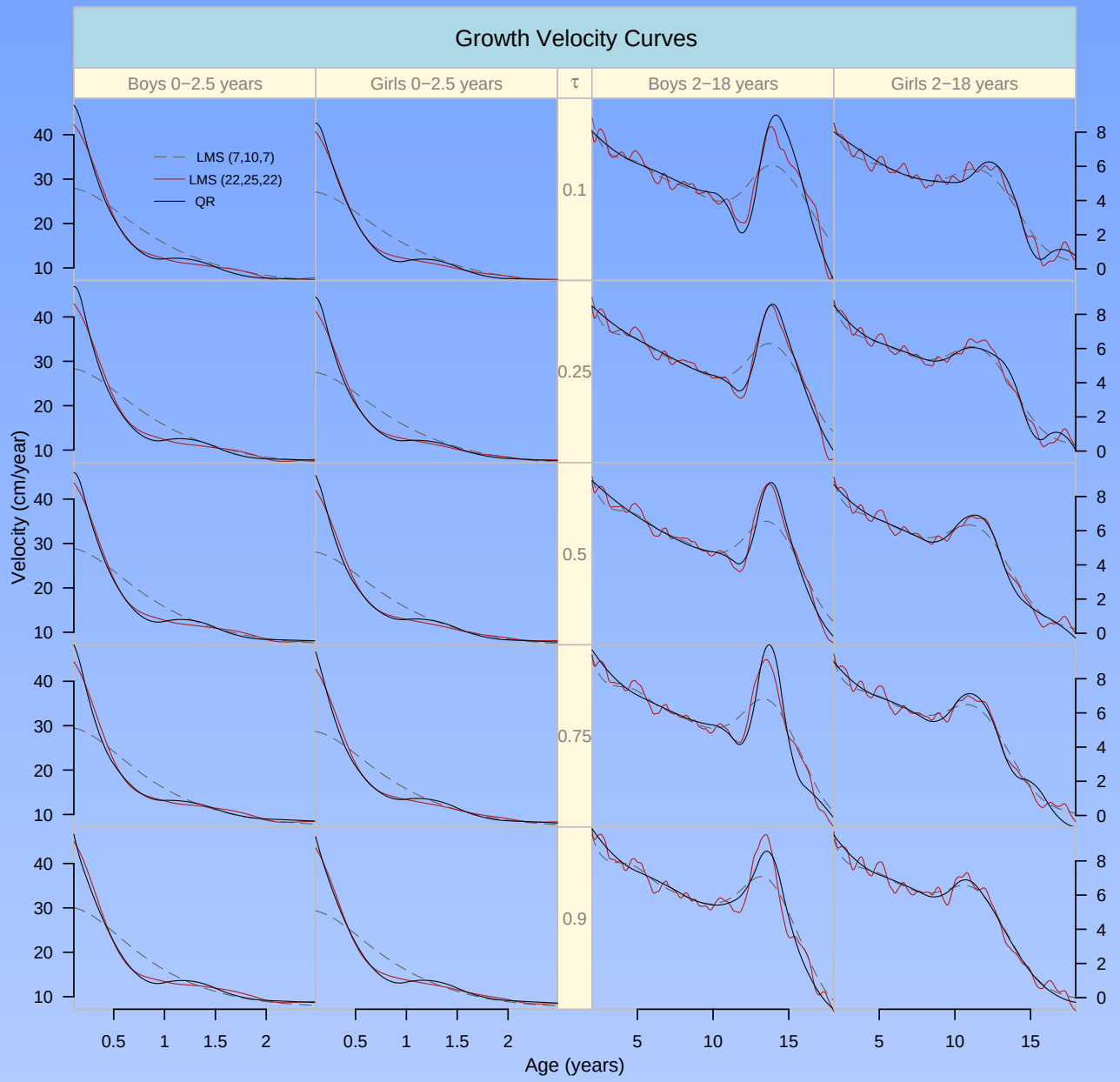
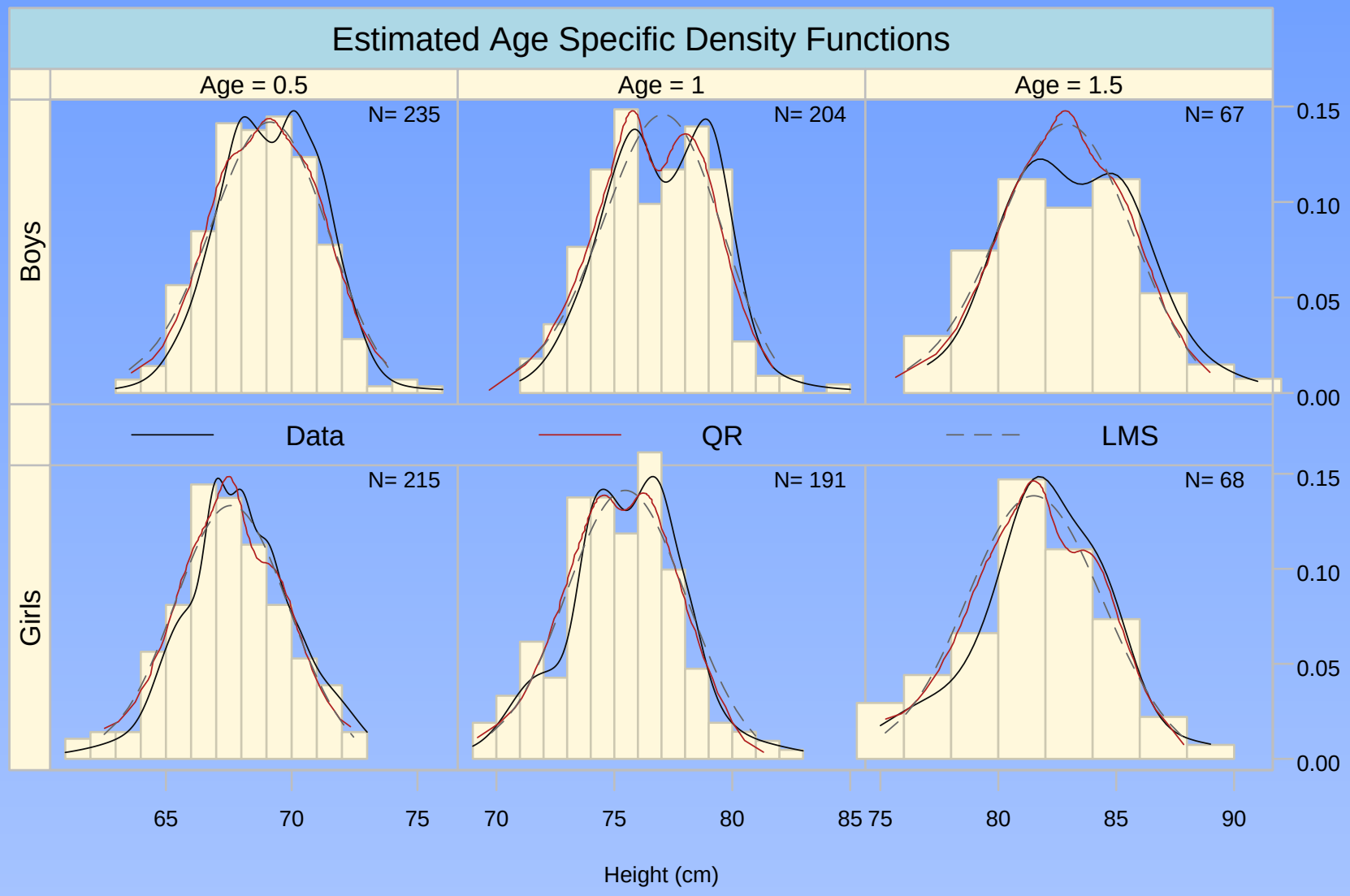
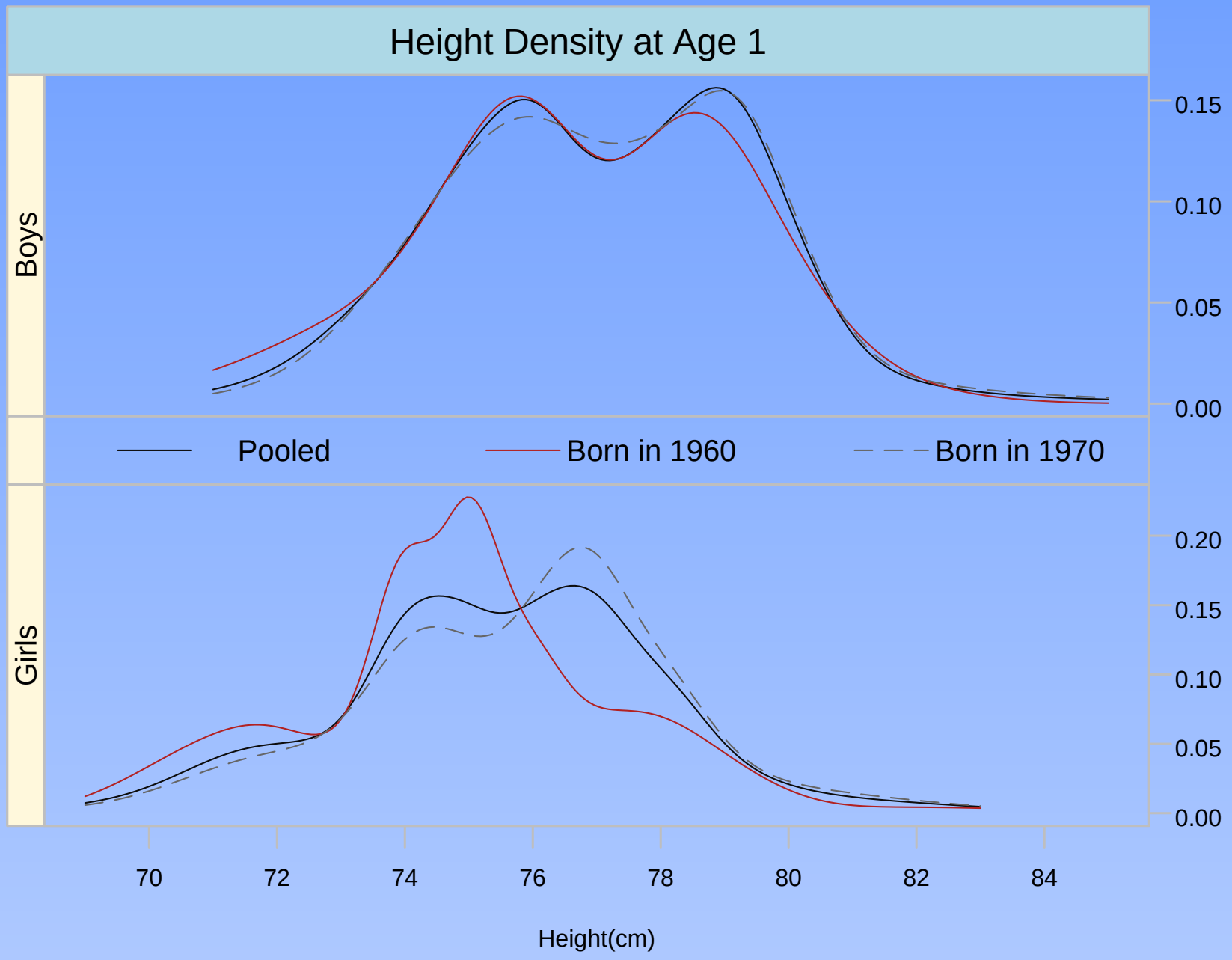

Growth Velocity Curves
Boys 0−2.5 years
Girls 0−2.5 years
τ
Boys 2−18 years
Girls 2−18 years
LMS (7,10,7)
LMS (22,25,22)
QR
0.1
0.25
0.5
0.75
0.9
Velocity (cm/year)
Age (years)


Estimated Age Specific Density Functions
Age = 0.5
Age = 1
Age = 1.5
Boys
Girls
N= 235
N= 204
N= 67
N= 215
N= 191
N= 68
Data
QR
LMS
0.15
0.10
0.05
0.00
65
70
75
70
75
80
85
75
80
85
90
Height (cm)


Height Density at Age 1
Boys
Girls
Pooled
Born in 1960
Born in 1970
0.15
0.10
0.05
0.00
0.20
0.15
0.10
0.05
0.00
70
72
74
76
78
80
82
84
Height(cm)

# Conditioning on Prior Growth

It is often important to condition not only on age, but also on prior growth and possibly on other covariates. Autoregressive models are natural, but complicated due to the irregular spacing of typical longitudinal measurements.

- Data: $\{Y_i(t_{i,j}) : \ j = 1, \ldots, J_i, \ i = 1, \ldots, n.\}$
- Model:

$$\begin{aligned} Q_{Y_i(t_{i,j})}(\tau \quad | \quad & t_{i,j}, Y_i(t_{i,j-1}), x_i) = g_\tau(t_{i,j}) \\ + \quad & [\alpha(\tau) + \beta(\tau)(t_{i,j} - t_{i,j-1})]Y_i(t_{i,j-1}) + x_i^\top \gamma(\tau). \end{aligned}$$

# AR Components of the Infant Conditional Growth Model

| $\tau$ | Boys | | | Girls | | |
|---|---|---|---|---|---|---|
| | $\hat{\alpha}(\tau)$ | $\hat{\beta}(\tau)$ | $\hat{\gamma}(\tau)$ | $\hat{\alpha}(\tau)$ | $\hat{\beta}(\tau)$ | $\hat{\gamma}(\tau)$ |
| 0.03 | 0.845 (0.020) | 0.147 (0.011) | 0.024 (0.011) | 0.809 (0.024) | 0.135 (0.011) | 0.042 (0.010) |
| 0.1 | 0.787 (0.020) | 0.159 (0.007) | 0.036 (0.007) | 0.757 (0.022) | 0.153 (0.007) | 0.054 (0.009) |
| 0.25 | 0.725 (0.019) | 0.170 (0.006) | 0.051 (0.009) | 0.685 (0.021) | 0.163 (0.006) | 0.061 (0.008) |
| 0.5 | 0.635 (0.025) | 0.173 (0.009) | 0.060 (0.013) | 0.612 (0.027) | 0.175 (0.008) | 0.070 (0.009) |
| 0.75 | 0.483 (0.029) | 0.187 (0.009) | 0.063 (0.017) | 0.457 (0.027) | 0.183 (0.012) | 0.094 (0.015) |
| 0.9 | 0.422 (0.024) | 0.213 (0.016) | 0.070 (0.017) | 0.411 (0.030) | 0.201 (0.015) | 0.100 (0.018) |
| 0.97 | 0.383 (0.024) | 0.214 (0.016) | 0.077 (0.018) | 0.400 (0.038) | 0.232 (0.024) | 0.086 (0.027) |

# AR Components of the Childrens' Conditional Growth Model

| $\tau$ | Boys | | | Girls | | |
|---|---|---|---|---|---|---|
| | $\hat{\alpha}(\tau)$ | $\hat{\beta}(\tau)$ | $\hat{\gamma}(\tau)$ | $\hat{\alpha}(\tau)$ | $\hat{\beta}(\tau)$ | $\hat{\gamma}(\tau)$ |
| 0.03 | 0.976 (0.010) | 0.036 (0.002) | 0.011 (0.013) | 0.993 (0.012) | 0.033 (0.002) | 0.006 (0.015) |
| 0.1 | 0.980 (0.005) | 0.039 (0.001) | 0.022 (0.007) | 0.989 (0.006) | 0.039 (0.001) | 0.008 (0.007) |
| 0.25 | 0.978 (0.006) | 0.042 (0.001) | 0.021 (0.006) | 0.986 (0.005) | 0.042 (0.001) | 0.019 (0.006) |
| 0.5 | 0.984 (0.004) | 0.045 (0.001) | 0.019 (0.004) | 0.984 (0.007) | 0.045 (0.001) | 0.022 (0.006) |
| 0.75 | 0.990 (0.004) | 0.047 (0.001) | 0.014 (0.006) | 0.985 (0.007) | 0.050 (0.001) | 0.016 (0.006) |
| 0.9 | 0.987 (0.009) | 0.049 (0.001) | 0.012 (0.009) | 0.984 (0.008) | 0.052 (0.001) | 0.002 (0.012) |
| 0.97 | 0.980 (0.014) | 0.050 (0.002) | 0.023 (0.015) | 0.982 (0.013) | 0.053 (0.001) | 0.021 (0.018) |

# Transformation to Normality

A presumed advantage of univariate (age-specific) transformation to normality is that once observations are transformed to univariate "Z-scores" they are automatically prepared to longitudinal autoregression:

$$Z_t = \alpha_0 + \alpha_1 Z_{t-1} + U_t$$

**Premise: Marginal Normality $\Rightarrow$ Joint Normality**

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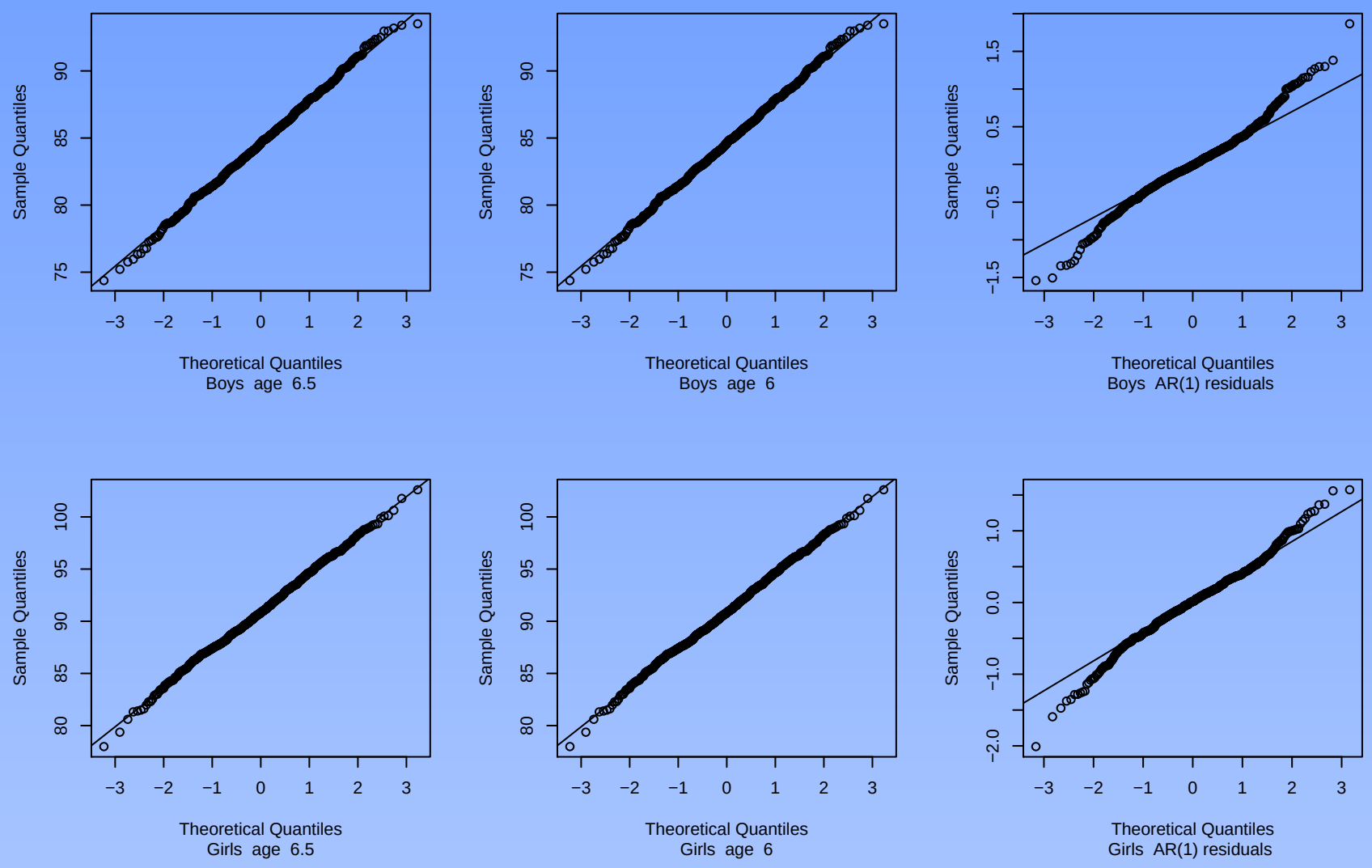
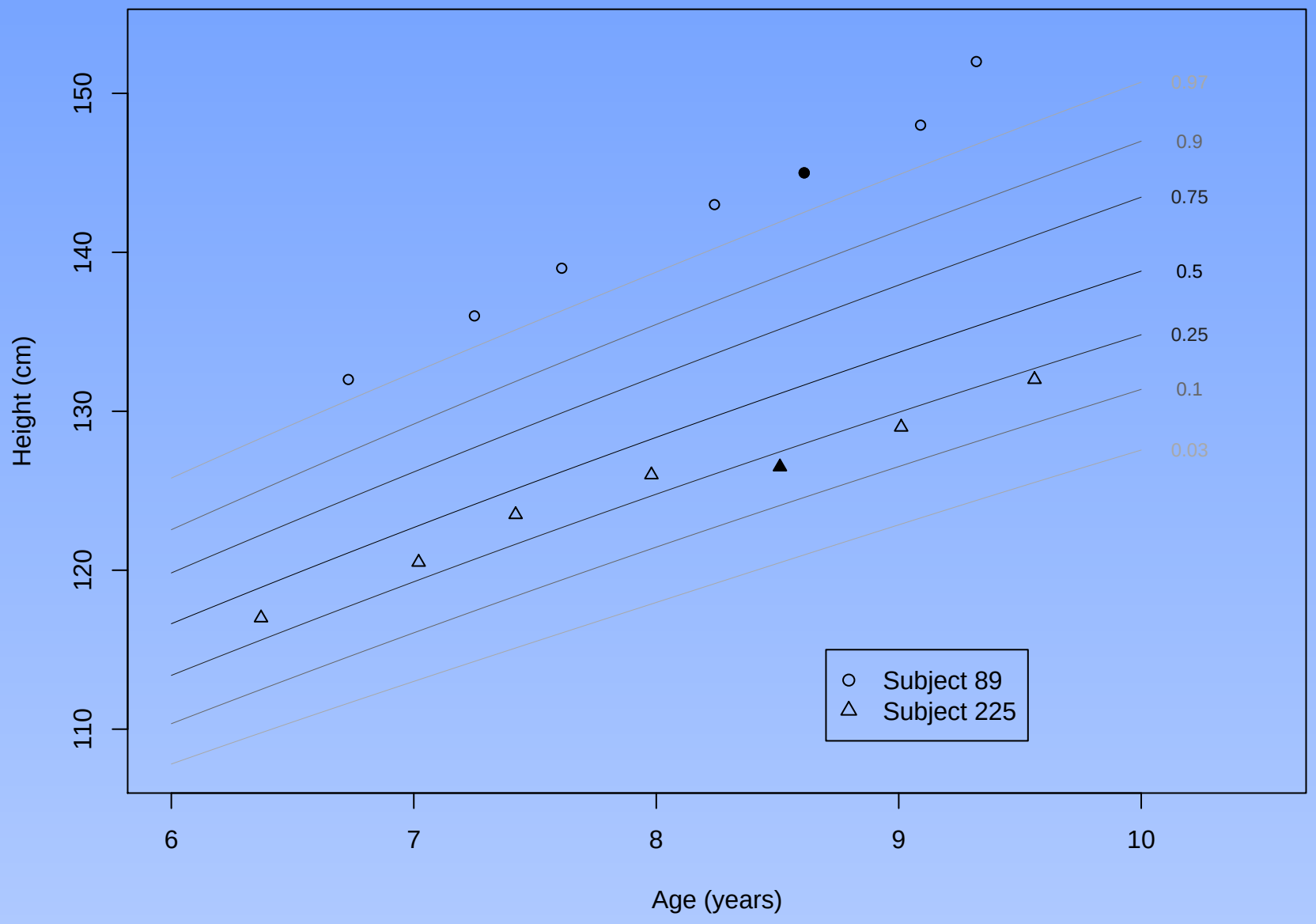
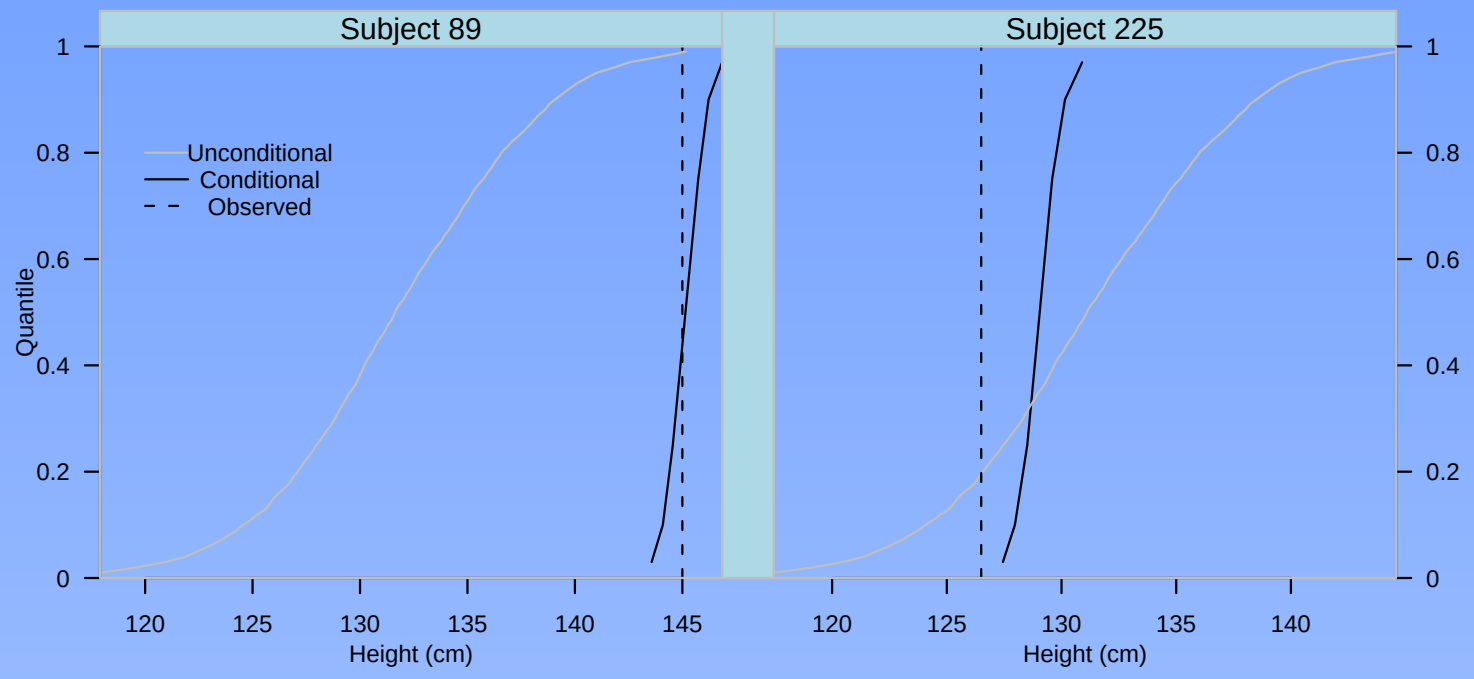
**Premise: Marginal Normality $\Rightarrow$ Joint Normality**

Of course we know it isn't true, but we also think we know that counterexamples are pathological, and don't occur in "nature."


Sample Quantiles
Theoretical Quantiles
Boys age 6.5
Boys age 6
Boys AR(1) residuals
Girls age 6.5
Girls age 6
Girls AR(1) residuals


Height (cm)
110
120
130
140
150
Age (years)
6
7
8
9
10
0.97
0.9
0.75
0.5
0.25
0.1
0.03
Subject 89
Subject 225


Subject 89
Subject 225
Unconditional
Conditional
Observed
Quantile
0
0.2
0.4
0.6
0.8
1
120
125
130
135
140
145
Height (cm)

# Conclusions

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- The flexibility of quantile regression methods exposes features of the data that are not easily observable with conventional parametric methods. *Even for height data.*

- Longitudinal data can be easily accomodated into the quantile regression framework by adding covariates, including the use of autoregressive effects for unequally spaced measurements.